RESUMPTION

Helix OTA — Session Resumption (canonical, always-current)

Field	Value
Revision	9
Last modified	2026-06-10T16:30:00Z
Status	active — the §11.4.131 single canonical out-of-the-box session-resumption file
Standard path	`docs/RESUMPTION.md` (this file — the fixed project-declared §11.4.131 entry point; do not move without a §11.4.66 operator decision)
Status summary	Point any fresh session at THIS file. It carries a SHORT one-line resume + a FULL block, the read-first handoff docs, exact live-state anchors, current PHASE + NEXT + terminal goal, and the binding constraints. Moment-valid for HEAD `ae6c2b7` on `main` (all 5 remotes aligned). **Overnight autonomous session 2026-06-10 (operator away-from-keyboard): EVERY achievable software tier RE-RUN fresh and GREEN, PLUS §11.4.50 determinism soaks (Go race ×5 + deep ×10, dashboard Vitest ×3) and a 30s/c=64 sustained loadtest (1.14M req, 38k rps, p99 7.2ms, 0 non-2xx) — all GREEN. Full §11.4.27 test-type matrix (unit/integration/e2e/full-automation/security/DDoS/scaling/chaos/stress/perf/UI/Challenges) re-confirmed. See `docs/qa/STABILITY_REPORT.md` Rev 2 + `docs/qa/20260610T16{20Z-overnight-revalidation,40Z-determinism-soak}/`.** Android bricks product-identical to proof (pins

Field	Value
	`1061015/8bb8d2f` unchanged), not re-run (§12.6/§11.4.6). BLOCKED (hardware): T2 Cuttlefish / GSI-A·B real-apply / T3 RK3588 → no release tag (§11.4.40 would be a bluff). Achievable+zero-risk queue is EXHAUSTED; remaining items are hardware-gated or correctly-deferred-as-risky (fabric scheduler = unwired §11.4.124).

SHORT — paste this first sentence into a fresh session

> Read `docs/research/main_specs/CONTINUATION.md` first, run `git fetch --all --prune`, then continue the Helix OTA autonomous loop toward the next validated-and-published version tag — current PHASE is "OVERNIGHT STABLE — every achievable tier RE-VERIFIED GREEN on this HEAD (see docs/qa/STABILITY_REPORT.md Rev 2); P4 Cuttlefish + GSI-A/B + T3 RK3588 remain hardware-BLOCKED so no release tag yet (§11.4.40); achievable+zero-risk queue EXHAUSTED"; HEAD is `ae6c2b7` on `main` (all 5 remotes aligned).

FULL — detailed resumption block

0. Read FIRST (in order)

1. `docs/research/main_specs/CONTINUATION.md` — the live work-state handoff (DONE / NEXT / locked decisions). **This is the primary handoff doc.**
2. `.remember/remember.md` — present but currently empty; no extra state there.
3. `docs/design/EMULATED_DEVICE_TESTING.md` — the tiered emulator-testing plan (Tier-1/2/3) now in flight.
4. Then run: `git fetch --all --prune` and reconcile `HEAD..@{u}` before any edit (§11.4.37 fetch-before-edit).

1. Exact live-state anchors (moment-valid 2026-06-10)

- **HEAD commit:** `ae6c2b7` — `test(overnight): §11.4.50 determinism soak + sustained loadtest` (on `main`, all 5 remotes aligned). Parent `8c93ec6` = overnight re-validation + doc-sync; grandparent `eb9e1c4`. **Build state: GREEN — comprehensively re-verified + determinism-soaked on this HEAD; see docs/qa/STABILITY_REPORT.md Rev 2.** NOTE (§11.4.6 fact-correction): `weasyprint` 66.0 + `pandoc` 3.9 ARE present on this host — the prior "no weasyprint/LaTeX → PDF gap" note is stale. PDF siblings ARE generated via the documented §11.4.65 pipeline pandoc <f>.md -s --metadata title=<base> -o <f>.html && weasyprint <f>.html <f>.pdf (NOTE: `scripts/export_docs.sh`'s OWN PDF path needs a LaTeX engine, which IS genuinely absent — use the weasyprint pipeline for PDF).

- **Wave 6 shipped this session (2026-06-10, on `main`, all 4 upstreams):**
  - `16b4b23` android gitlink → ota-android-agent `26bd0a2` (**`:android` AAR now builds** — root-caused to an unpinned `kotlin.android` plugin, NOT MPP; `:core` 47/0 green).
  - `12087bc` **emulator failure→recall(forward-fix)→recovery e2e** (`TestRecallRecoveryE2E` PASS +`-race`; podman shell variant `tests/emulator/tier1_recall_recovery_e2e.sh` syntax-clean, **conductor must run it live**).
  - `19e1346` **dashboard §11.4.135 regression guard** for the DeviceStatus.health-object crash (Vitest 58, RED/GREEN polarity).
  - `3c85b14` **security probes** recall+telemetry 28/0 (surfaced a STRONGER trust boundary — strict `bindJSON` rejects injected keys 400).
  - `88bd2c2` **recall lifecycle e2e** live 35/0 (`tests/e2e/recall_lifecycle.sh`).
  - `5123e67` **HelixQA bluff audit + real bank runner** (`tools/helixqa/run_bank.sh`): VERDICT **HelixQA is NOT a bluff** — canonical engine gates on dispatch-exit-0 + non-empty evidence ledger + catches its own negation; the real gap was an INCORPORATION gap (§11.4.27, never wired as a submodule / bank never machine-executed). Runner: dry-run 10/0/0 + self-test PASS. Audit: `docs/research/helixqa_bluff_audit/REPORT.md`.
  - `71be1cd` **per-event telemetry `duration_ms`+`bytes_transferred` end-to-end** (closed the parked WIDEN row-4): ota-protocol gitlink → `3d360ab` (additive, validation rejects negative), emulator emits → server ingest → store (memory+pgx) → read view; `TestEmulatorTelemetryFieldsEndToEnd` exact-value PASS + real-Postgres parity (`TestPostgresRepositoryContract_Integration`, podman PG, 0 skips).
  - `8a03a63`+`de87e21` **emulation test-fabric research + full design** (`docs/research/emulation_infra/REPORT.md` cited; `docs/design/emulation_fabric/{DESIGN,ROADMAP,TEST_COVERAGE_PLAN}.md`+`SCHEMA.sql`) — tiers T0–T3/Tfw/Tcp, extends containers submodule, honest macOS-M3 KVM gaps.
- **Queue done this session (all pushed, all 4 upstreams):** ✅ HelixQA LIVE full-bank **10/0** (`d5fcf4a` — fixed run_bank LIVE-mode <pw> substitution + free-port isolation); ✅ podman recall→recovery e2e **7/7** (`71c3f77`); ✅ **HelixQA wired as a submodule** at `submodules/helixqa` pinned latest `fe17c08`, gate Part C PASS, install_upstreams done (`d488e6e`, §11.4.27); ✅ **emulation P1 T1-tier AVD-HVF boot smoke** arm64-v8a booted+accel-proven (`5d6b3ae`).
- **Wave 7 (5-agent parallel, all pushed):** ✅ **fabric target-registry** project-side Go (`c709c92` — server/internal/fabric + store memory+pgx, exclusive-lease §11.4.119 + non-empty-evidence §11.4.69, real-PG integration 0 skips, §1.1 mutation); ✅ **on-device AVD instrumentation test** → ota-android-agent `1061015` (`8266355` gitlink; OK 5 tests 3x; UpdateEngine-present-but-degrades-to-Failed FACT); ✅ **P2 Tfw QEMU smoke** authored, honest SKIP (qemu absent — `brew install qemu` to activate) (`504c92b`); ✅ **helix-deps.yaml §11.4.31** + native test_cases bank (`607e7c3`); ✅ spec_impl_alignment row-4 RESOLVED + coverage_ledger Rev2 (`7a7cb48`).
- **Overnight stability hardening (2026-06-10, all pushed):** ✅ **P2 QEMU firmware tier ACTIVATED** — `brew install qemu` (11.0.1), tier_fw_qemu_smoke.sh real edk2 UEFI boot PASS (`93ac0a3`); ✅ **run_bank –self-test wired into the pre-build gate** (`60da61a`, §1.1 every build); ✅ **Challenges submodule incorporated** at submodules/challenges (`438057d`,

§11.4.27, gate PASS, helix-deps 11 deps); ✅ **comprehensive stability sweep ALL GREEN** on HEAD 438057d (Go all-tiers+race, pgx integration, meta-test, dashboard 58, HelixQA LIVE bank 10/0, podman full-lifecycle + fleet 5/5 + recall-recovery, QEMU tier, AVD boot + on-device, submodule cores) — see `docs/qa/STABILITY_REPORT.md`. DEFERRED zero-risk: fabric scheduler P3 (would be unwired §11.4.124), HelixQA in-tree compile (../containers layout, HelixQA-side). [CORRECTED 2026-06-10 overnight: the prior "PDF doc siblings (no weasyprint/LaTeX)" deferral was stale — weasyprint 66.0 IS present; PDF siblings for the handoff docs ARE now generated via pandoc+weasyprint; only a full-corpus PDF backfill remains as optional follow-up.]
- **Remaining NEXT (real PWUs):** (1) emulation **P4 Cuttlefish** (Linux+KVM) for real `update_engine` A/B + the **GSI-A/B real-apply** question (REPORT §2.2/§8 — on a google_apis AVD UpdateEngine is present but unusable for a non-system caller); (2) **P2 QEMU** activate (`brew install qemu` → tier_fw_qemu_smoke.sh goes green); (3) **P3 fabric scheduler** wiring on top of the registry (Nomad/LAVA backend); (4) wire HelixQA's missing `challenges` submodule (its go.mod `replace digital.vasic.challenges` points at an absent `submodules/challenges`); (5) HelixQA: `run_bank.sh --self-test` into the pre-build gate, Dispatcher→autonomous-runner (§11.4.124); (6) OPTIONAL full-corpus PDF backfill: weasyprint 66.0 + pandoc 3.9 ARE present, and the handoff docs' .pdf siblings are now generated via `pandoc -s … && weasyprint`; a one-shot pass to (re)generate .pdf siblings across the ENTIRE docs corpus (not just handoff docs) is the only remaining §11.4.65 PDF item — low-priority, non-blocking.
- **Wave 5 shipped (on `main`):** `2391cb6` **full OTA lifecycle + multi-device fleet PROVEN on podman** (single device 1.0.0→1.1.0 with the complete download→success telemetry accepted; fleet 5/5 → 1.0.0→1.2.0; evidence `docs/qa/20260610T11191*/` + `...111928Z-fleet/`); `f5a3428` N-device scaling (50 concurrent, ~1580 ops/sec, 0 err, -race); `4658125` dashboard Fleet-detail + a real product bug fix (DeviceStatus.health object). **The emulator now fully runs+tests the codebase end-to-end on podman — the prior "honest gap" is closed.**
- **NEXT non-blocked candidates:** richer telemetry ingest fields (duration_ms/ bytes_transferred — needs the device to send them); a rollback/recall lifecycle e2e on the emulator (drive a failure event → recall forward-fix); migrate the dashboard health-bug to a Fixed.md ATM entry. **BLOCKED:** ATM-003 Tier-2 (Linux+KVM Cuttlefish, design ready in docs/design/ CUTTLEFISH_TIER2.md) + ATM-004 Tier-3 (RK3588 hardware).
- **Waves 3–4 also shipped (on `main`):** `3c57867` closed both protocol gaps (UpdateAvailable.deployment_id [ota-protocol→7920842] + GET / deployments); `8c0521d` Tier-1 podman container e2e (PROVEN: control-plane container boots, ota-device-emu container runs the real round-trip — evidence docs/qa/20260610T105306Z/); `5d4920e` dashboard ArtifactUpload + populated-detail (Vitest 50, Playwright 20); submodule pins bumped — constitution `ba0f702`, ota-rollout-engine `7a90912`, ota-artifact-validator `77c6b48`, containers `845ad45`. Tracker live at docs/Issues.md (ATM-003/004 Operator-blocked: Tier-2 Cuttlefish-on-Linux-KVM / Tier-3 RK3588 hardware) + docs/ Fixed.md.
- **Just-shipped this session (on `main`, pushed all 4 upstreams):**
  - `50ef5c6` — `feat(api): per-device telemetry filters + group/ members pagination` (OpenAPI synced, redocly-clean).
  - `b0b8ee2` — `chore(dashboard): sync API client to new pagination/filter params` (dashboard client lockstep).

    - 7dc3334 — `feat(emulator): Tier-1 Go OTA device-emulator + resilience` (`server/internal/deviceemu` + `cmd/ota-device-emu`; surfaced the telemetry deployment_id protocol gap).
    - fa571b8 — `test(dashboard): comprehensive UI testing system` (Vitest 43 + Playwright 17 + a11y).
    - a839220 — `test(qa): autonomous e2e (50/50) + security (39/39) + HelixQA bank`.
    - **`containers` submodule** advanced to `845ad45` — `feat(emulator): ota-device-emu runtime image + on-demand boot recipe` (pushed to its github origin); parent gitlink bumped in this commit.
- **Branch:** `main`. All work merges to `main`; commit + push fan out to all upstreams.
- **Upstreams (4):** `github` (`git@github.com:HelixDevelopment/helix_ota.git`, also `origin` fetch), `gitlab` (`git@gitlab.com:helixdevelopment1/helix_ota.git`), `gitflic` (`git@gitflic.ru:helixdevelopment/helix_ota.git`), `gitverse` (`git@gitverse.ru:helixdevelopment/helix_ota.git`). `origin` push fans out to all four.
- **No in-flight background PIDs / detached pushes known at handoff time** — verify with `git status` + check `qa-results/push_failures/` per §11.4.88 before assuming clean.
- **Container runtime on this host:** `podman` (the pgx integration + dev stack boot via the `containers` submodule on podman). **No host QEMU / no AVD / no nested KVM on this Apple-Silicon host** — see binding constraints.

2. Current PHASE + immediate NEXT + terminal goal

- **PHASE:** Emulator-driven device testing — **Tier-1 in progress** (a podman container running a Go `ota-device-emulator` that speaks the real `ota-protocol` to the control plane). See `docs/design/EMULATED_DEVICE_TESTING.md`.
- **Immediate NEXT (priority order):**
    1. Tier-1 emulator: stand up the podman `ota-device-emulator` exercising register → update-check → telemetry → delta → rollout → recall against a live control plane, capturing real evidence under `docs/qa/<run-id>/`.
    2. Remaining NEXT-wave items (hardware/ingest-gated): device-side TUF (gomobile-go-tuf/v2 — gated on real RK3588 `.so`/JNI measurement); device payload-apply integration (`DeltaApplyDecision` → update_engine, needs a real device); row-4 richer telemetry fields (`duration_ms`/`bytes_transferred` — blocked on UNVERIFIED ingest).
- **Terminal goal (loop stop condition A, §11.4.126):** a new fully-validated-and-verified version (tag) created AND published across all owned submodules + main repo to all 4 upstreams.

3. Binding constraints (do not violate)

- **Anti-bluff covenant §11.4** — every PASS carries positive captured evidence; metadata-only / config-only / absence-of-error / grep-without-runtime PASS forbidden. Tests AND HelixQA Challenges bound equally.
- **Absolute no-force-push §11.4.113** — force-push (`--force`, `--force-with-lease`, `+<ref>`) is STRICTLY forbidden on every repo/submodule. Integrate via fetch → merge-onto-latest-main → fast-forward push.

- **podman-only on this host** — use the `containers` submodule (`vasic-digital/containers`, §11.4.76) for any containerized workload; never host-direct emulator/`adb`/`qemu` (§11.4.109 guard).
- **Tier-2 Android A/B is host-gated** — real `update_engine` A/B + AVB/dm-verity auto-rollback needs Cuttlefish on a Linux + nested-KVM box; the Apple-Silicon `applehv` host cannot run it. NOT structurally impossible (§11.4.112) — host/hardware-gated only.
- **Exact naming/versions:** Go module `github.com/HelixDevelopment/helix_ota/server`; submodules under `submodules/`; constitution submodule at `constitution/`. Toolchain: Go 1.26.2, Gradle 9.5 + Kotlin 2.3.20 (AGP 8.5.2 + plain `kotlin.android` builds; Kotlin MPP does NOT). No LaTeX/drawio/plantuml/graphviz/docker on host (so `export_docs.sh`'s LaTeX-based PDF path is unavailable) — BUT PDF siblings ARE produced via `pandoc -s` + `weasyprint` 66.0 (both present); use that pipeline for §11.4.65 PDF.
- **Commit/push discipline §2.1 + §11.4.88** — commit + push to ALL 4 upstreams; pushes may run detached.

How a fresh session resumes from this file alone

Given only this file's path, a new agent: reads §0 handoff docs, runs `git fetch --all`, confirms HEAD against §1, picks up the §2 PHASE/NEXT, and works under the §3 constraints toward the terminal goal — zero additional context required.